Lecture for 11th week

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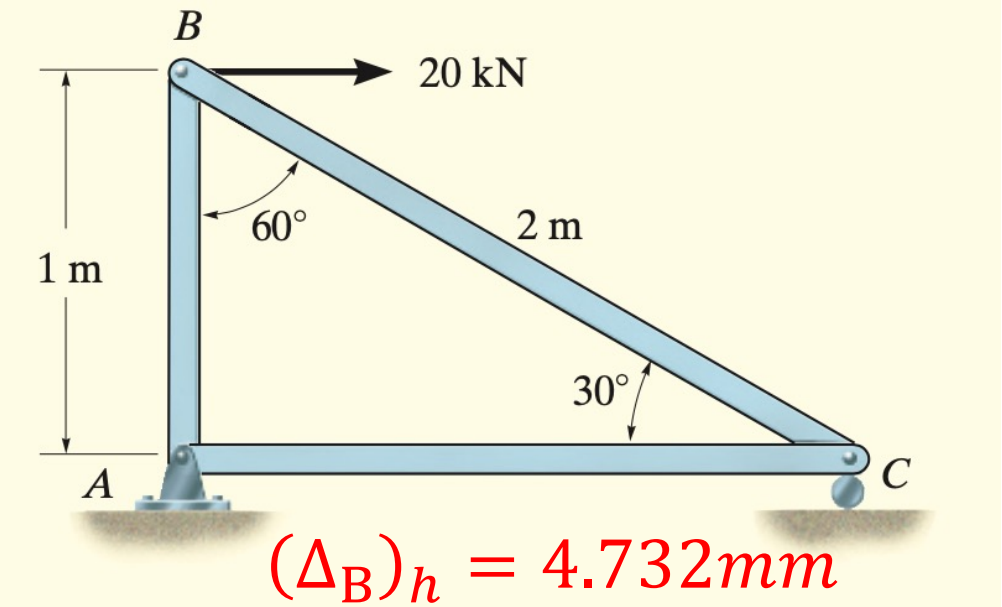
Mechanical Engineering & Robotics,

Guangdong Technion - Israel Institute of Technology

26 May 2026

1. Examples-1

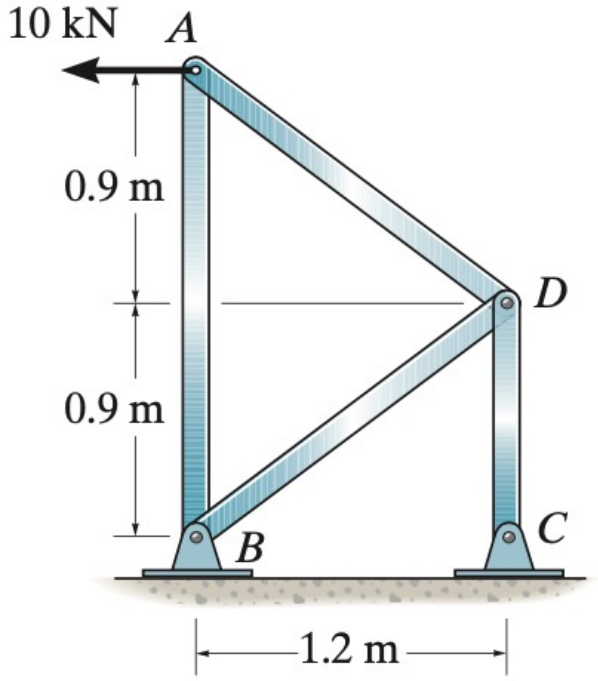
The three-bar truss is subjected to a horizontal force of 20 kN. If the cross-sectional area of each member is 100 mm^2, determine the horizontal displacement at point B. $E = 200$ GPa.



$$(\Delta_B)_h = 4.732mm$$

1. Examples-2

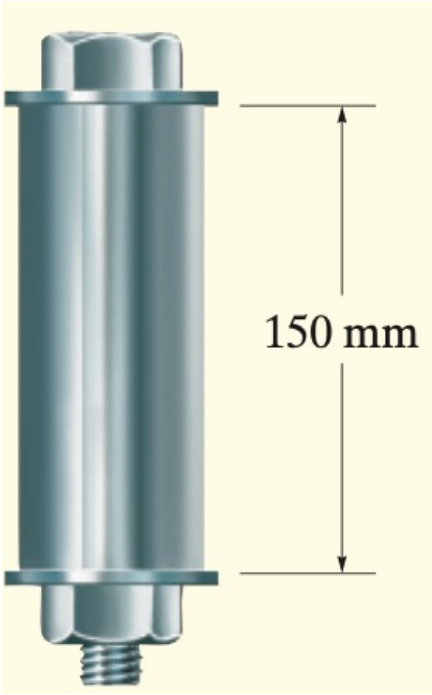
Determine the horizontal displacement of joint A. Each bar is made of A-36 steel (E = 200GPa) and has a cross-sectional area of 950 mm2.



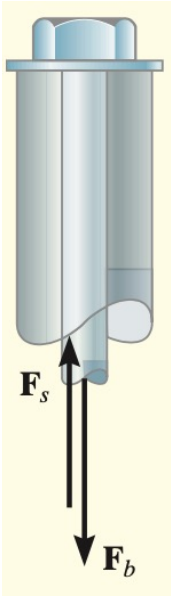
$$(\Delta_A)_h = 0.407mm$$

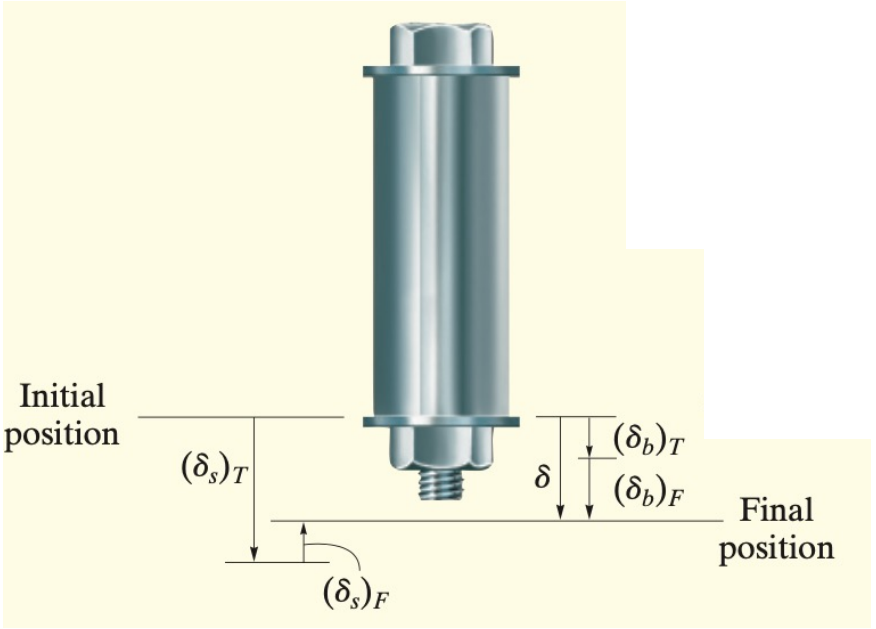
1. Examples-3

A 2014-T6 aluminum ($\alpha = 23*(10^{-6})/°C$, E=93.1GPa) tube having a cross-sectional area of 600 mm^2 is used as a sleeve for an A-36 steel ($\alpha = 12*(10^{-6})/°C$, E=200GPa) bolt having a cross-sectional area of 400 mm^2. When the temperature is $T_1 = 15°C$, the nut holds the assembly in a snug position such that the axial force in the bolt is negligible. If the temperature increases to $T_2 = 80°C$, determine the force in the bolt and sleeve.



1. Examples-3

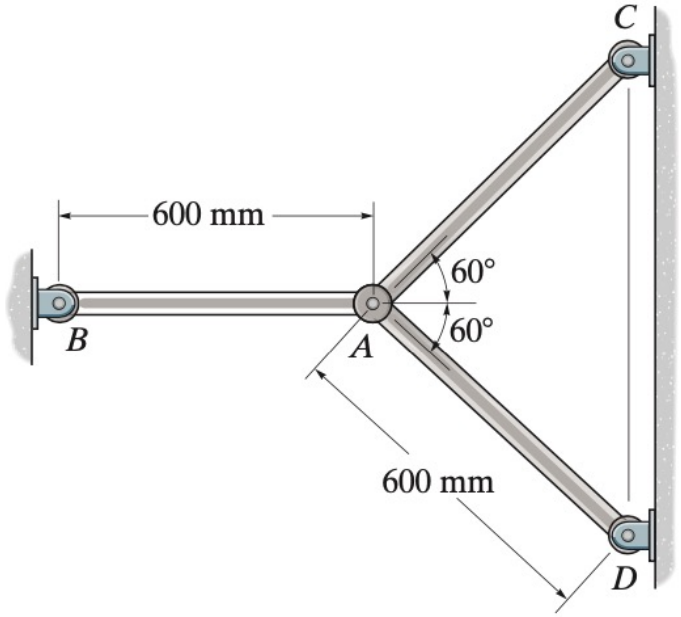




$$F_s = Fb$$
$$= 20.3kN$$

1. Examples-4

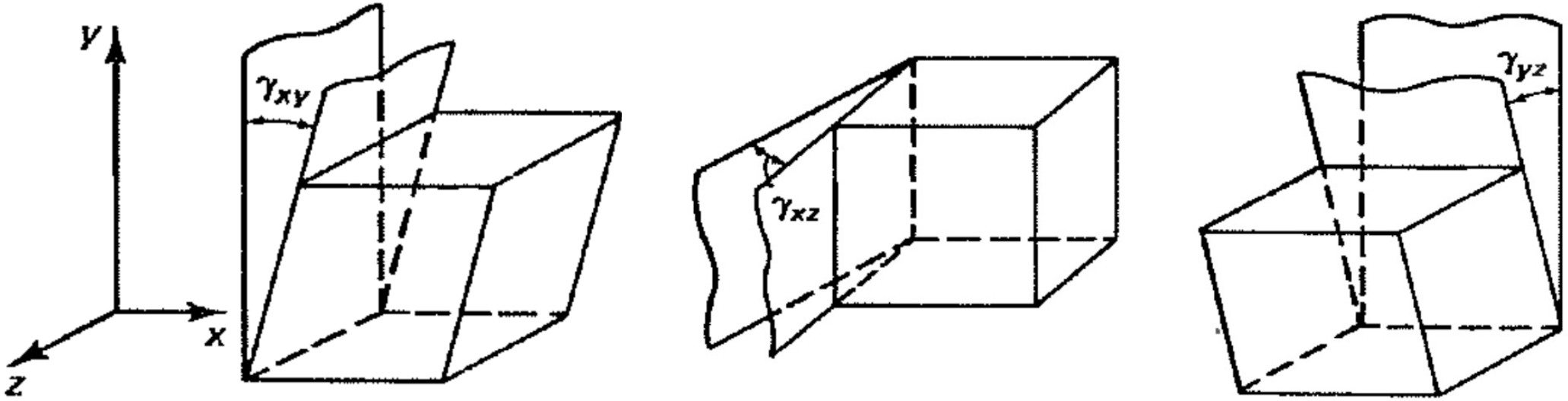
The rods each have the same 25mm diameter and 600mm length. If they are made of A992 steel ($\alpha = 23*(10^{-6})/°C$, E=200GPa), determine the forces developed in each rod when the temperature increases to 50° C.



$$F_{AB} = FAC = FAD = 58.9kN(C)$$

2. Introduction

Shear Strain γ : the change in the initial right angle between any two imaginary planes.

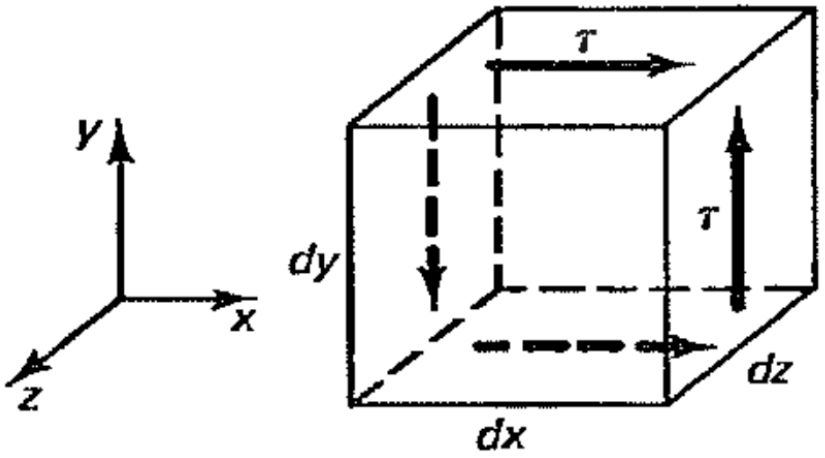


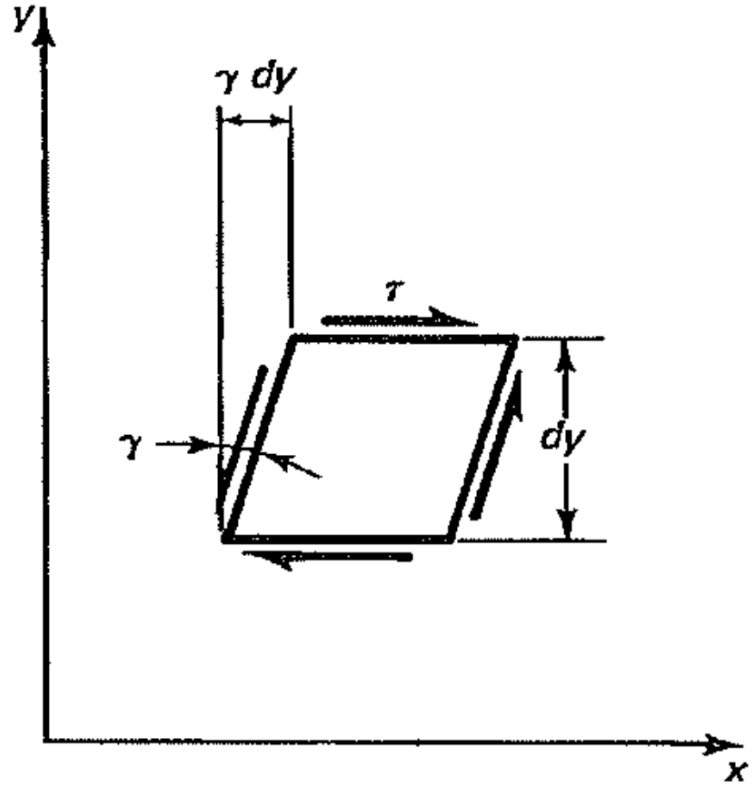
possible shear deformation of an element

2. Introduction

Extension of Hooke's law for shear stress and strain

$$\tau = G\gamma$$





2. Introduction

Shear stresses on mutually perpendicular planes are equal.

